Grade 11 - Term 3
C1–C4 Practice Exercises and Solutions

Contents

Evaluated criteria

C1 Characterises continuous random variables by distinguishing them from discrete variables and interpreting their probability density function, cumulative distribution function, and support in economic contexts.

C2 Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas.

C3 Applies circular-arc probability density functions to calculate interval probabilities using the geometry of circles and circular segments.

C4 Applies the exponential distribution to constant-rate waiting-time situations and assesses its suitability.

1 Identifying Probability Distributions from Contexts [C1-5]

A variety of Colombian organizations use probability models to represent uncertain quantities and events. In each situation below, a random variable is defined from the information provided. The appropriate probability distribution must be identified from the characteristics of the situation.

Questions/Tasks.

C1 For each context, characterize the random variable and identify the appropriate probability distribution. Briefly justify your choice based on the features described. **[5]**

Context 1. A payment platform monitors one scheduled electronic transfer. By the end of processing, it is either completed successfully or not completed. The variable X_1 records the outcome to help estimate verification and support needs.

Context 2. An agricultural technology company monitors the daily temperature variation inside a refrigerated storage unit. The variation X_2 ranges from $-2°\text{C}$ to $2°\text{C}$, with all values in this interval considered equally likely. The company uses this model to set an operating margin.

Context 3. A streaming platform sends the same offer to exactly 80 randomly selected eligible users. Each user either subscribes or does not. The variable X_3 counts the number of subscribers to estimate the server capacity needed for the promotion.

Context 4. A Bogotá parking company records the number of vehicle breakdowns during a fixed one-week period. Comparable records show that breakdowns occur independently and without a predictable timetable. The variable X_4 counts the breakdowns to help schedule maintenance staff.

Context 5. An online marketplace studies the delivery time X_5 for a particular type of order. Delivery times range from 20 to 80 minutes, with deliveries around 45 minutes occurring most frequently. They become progressively less common toward either extreme. The company uses this model to estimate delivery times.

Context 6. A quality-control laboratory tests electronic components until the first defective component is found. Components are tested independently, and testing stops after the first defect. The variable X_6 records the number of components tested, including the defective one, to estimate inspection costs.

Context 7. A cloud-computing provider studies storage usage X_7 among small business customers. Usage ranges from 0 to 50 gigabytes and becomes steadily more common toward the upper limit, without reaching a peak before 50 GB. The provider uses this model to plan capacity.

Context 8. A retailer sends a promotional message to exactly 120 customers selected using the same procedure. Each customer either makes a purchase or does not. The variable X_8 counts the number of purchases to estimate the inventory needed for the campaign.

Context 9. A coffee exporter measures the moisture content X_9 of coffee bags under usual storage conditions. Measurements cluster around 11

Context 10. A food-delivery company records the number of customer cancellations during a fixed three-hour dinner period in one delivery zone. Comparable evenings show cancellations occurring independently without a recurring timetable. The variable X_{10} counts the cancellations to estimate customer-service needs.

C1 - Distribution identification: Solutions [10]

The appropriate distribution for each context is determined by whether the variable is discrete or continuous and by the defining features of the situation:

1. **Context 1 — Discrete, Bernoulli.** The transfer is either completed or not completed.
2. **Context 2 — Continuous, Uniform.** All values within a fixed interval are equally likely.
3. **Context 3 — Discrete, Binomial.** The variable counts subscriptions among 80 comparable users.
4. **Context 4 — Discrete, Poisson.** The variable counts independent incidents occurring during a fixed period.

5. **Context 5 — Continuous, Triangular.** The variable has a minimum, maximum, and most likely value.
6. **Context 6 — Discrete, Geometric.** The variable counts trials until the first defective component is found.
7. **Context 7 — Continuous, Linear.** The density increases steadily toward the upper limit.
8. **Context 8 — Discrete, Binomial.** The variable counts purchases among 120 comparable customers.
9. **Context 9 — Continuous, Normal.** Values cluster symmetrically around a central mean.
10. **Context 10 — Discrete, Poisson.** The variable counts independent cancellations during a fixe

2 Delivery Time for a Local Grocery Order [C2-4]

A Colombian grocery-delivery company studies the additional time required to deliver orders during a busy evening period. For a particular delivery zone, the added delivery time X can range from 0 to 10 minutes. A delay of any length within this interval is possible, and shorter delays become steadily more common. Historical data indicate that no additional delay is 4 times as common as a 10-minute delay.

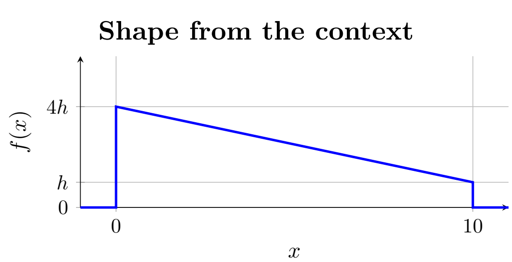
Questions/Tasks.

C2 Determine the complete probability density function for the added delivery time. Then determine the probability that the added delivery time exceeds 7 minutes, and the probability that it is between 3 and 7 minutes. **[4]**

C2 - Linear probability distribution [4]

2.1 Support and Shape

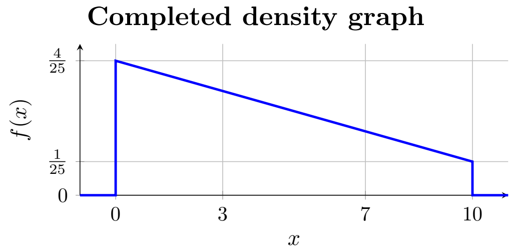
The random variable X has support $[0, 10]$. The context describes a steadily decreasing straight-line density whose height at 0 is four times its height at 10. Write the endpoint heights as $4h$ and h. The density is zero outside the support.



2.2 Total Area = 1

The total area under a probability density function must equal 1. Therefore, the trapezoid formed by the density has area 1:

$$1 = \frac{1}{2}(10)(4h + h) = 25h, \qquad h = \frac{1}{25}, \qquad 4h = \frac{4}{25}.$$



2.3 Straight Line Rule

The density is a straight line passing through the two endpoint points

$$(x_1, y_1) \quad \text{and} \quad (x_2, y_2). \quad \longrightarrow \quad m = \frac{y_2 - y_1}{x_2 - x_1}.$$

$$\left(0, \frac{4}{25}\right) \quad \text{and} \quad \left(10, \frac{1}{25}\right). \quad \longrightarrow \quad m = \frac{\frac{1}{25} - \frac{4}{25}}{10 - 0} = -\frac{3}{250},$$

and the y-intercept is

$$(x_1, y_1) \quad \text{and} \quad (x_2, y_2). \quad \longrightarrow \quad b = y_1 - mx_1 = y_2 - mx_2.$$

$$\left(0, \frac{4}{25}\right) \quad \text{and} \quad \left(10, \frac{1}{25}\right). \quad \longrightarrow \quad b = \frac{4}{25} - \left(-\frac{3}{250}\right)(0) = \frac{1}{25} - \left(-\frac{3}{250}\right)(10) = \frac{4}{25}.$$

Therefore, the probability density function is

$$f_X(x) = \frac{40 - 3x}{250}$$

on the stated interval.

2.4 Complete pdf

Including the values outside the possible range of added delivery times, the complete probability density function is

$$f_X(x) = \begin{cases} \dfrac{40 - 3x}{250}, & 0 \leq x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

2.5 Probability of More Than 7 Minutes

The company wants to know the chance that an order takes more than 7 additional minutes to deliver. This corresponds to the area under the density from 7 to the maximum possible delay of 10 minutes:

$$P(X > 7) = \frac{1}{2}(10 - 7)\left(f_X(7) + f_X(10)\right).$$

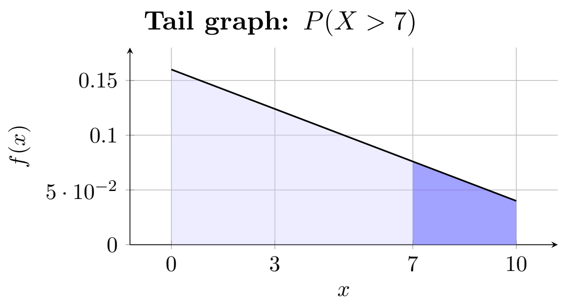
Since

$$f_X(7) = \frac{19}{250} \quad \text{and} \quad f_X(10) = \frac{1}{25},$$

$$P(X > 7) = \frac{1}{2}(3)\left(\frac{19}{250} + \frac{1}{25}\right) = \frac{9}{100}.$$

Thus, the probability of more than 7 minutes of additional delivery time is

$$\boxed{P(X > 7) = 0.09}.$$



2.6 Probability Between 3 and 7 Minutes

The company also wants to know the chance that the added delivery time falls between 3 and 7 minutes. This corresponds to the area under the density between these two times:

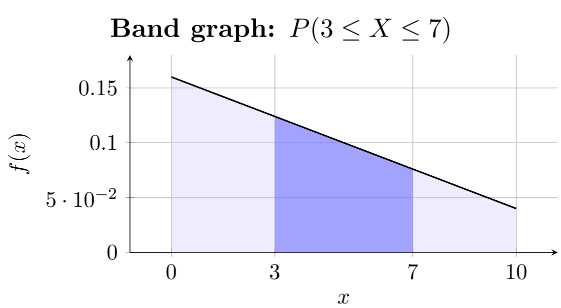
$$P(3 \le X \le 7) = \frac{1}{2}(7-3)\left(f_X(3) + f_X(7)\right).$$

Since

$$f_X(3) = \frac{31}{250} \qquad \text{and} \qquad f_X(7) = \frac{19}{250},$$
$$P(3 \le X \le 7) = \frac{1}{2}(4)\left(\frac{31}{250} + \frac{19}{250}\right) = \frac{2}{5}.$$

Thus, the probability of between 3 and 7 minutes of additional delivery time is

$$\boxed{P(3 \le X \le 7) = 0.40}.$$



3 Daily Commute Time for a Bogotá Apprentice [C2-4]

An apprentice records the time spent travelling to a training programme each day. Commute times range from 1 to 9 hours. Times become steadily more common from 1 hour up to a most common time of 4 hours, then steadily less common through 9 hours; there are no commute times right at either limit.

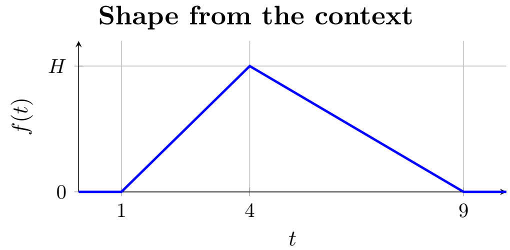
Questions/Tasks.

C2 Determine the complete probability density function for the commute time. Then determine the probability that the commute takes more than 6 hours, and the probability that it takes between 3 and 7 hours. **[4]**

C2 - Triangular probability distribution **[4]**

3.1 Support and Shape

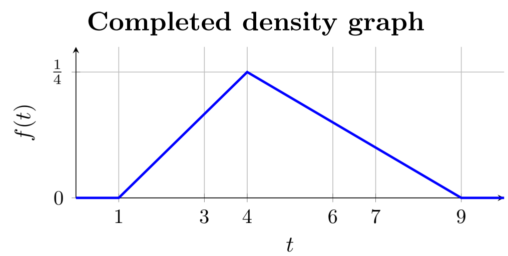
The random variable T has support $[1, 9]$. The context supplies zero height at both endpoints, a steady rise to the most common value at $t = 4$, and a steady fall thereafter. Denote the unknown peak height by H; the density is zero outside the support.



3.2 Total Area = 1

The triangle must have area 1:

$$1 = \frac{1}{2}(9-1)H = 4H, \qquad H = \frac{1}{4}.$$



3.3 Straight Line Rule

For the rising segment, the density is a straight line passing through the two endpoint points

$$(x_1, y_1) \qquad \text{and} \qquad (x_2, y_2)\,. \qquad \longrightarrow \qquad m = \frac{y_2 - y_1}{x_2 - x_1}.$$

$$(1, 0) \qquad \text{and} \qquad \left(4, \frac{1}{4}\right). \qquad \longrightarrow \qquad m_1 = \frac{\frac{1}{4} - 0}{4 - 1} = \frac{1}{12}.$$

The y-intercept is

$$(x_1, y_1) \qquad \text{and} \qquad (x_2, y_2)\,. \qquad \longrightarrow \qquad b = y_1 - mx_1 = y_2 - mx_2.$$

$$(1, 0) \qquad \text{and} \qquad \left(4, \frac{1}{4}\right). \qquad \longrightarrow \qquad b_1 = 0 - \frac{1}{12}(1) = \frac{1}{4} - \frac{1}{12}(4) = -\frac{1}{12}.$$

Therefore, the equation of the rising segment is

$$y = m_1 t + b_1 = \frac{t-1}{12}.$$

For the falling segment, the density is a straight line passing through the two endpoint points

$$(x_1, y_1) \qquad \text{and} \qquad (x_2, y_2)\,. \qquad \longrightarrow \qquad m = \frac{y_2 - y_1}{x_2 - x_1}.$$

$$\left(4, \frac{1}{4}\right) \qquad \text{and} \qquad (9, 0). \qquad \longrightarrow \qquad m_2 = \frac{0 - \frac{1}{4}}{9 - 4} = -\frac{1}{20}.$$

The y-intercept is

$$(x_1, y_1) \qquad \text{and} \qquad (x_2, y_2). \qquad \longrightarrow \qquad b = y_1 - mx_1 = y_2 - mx_2.$$

$$\left(4, \frac{1}{4}\right) \qquad \text{and} \qquad (9, 0). \qquad \longrightarrow \qquad b_2 = \frac{1}{4} - \left(-\frac{1}{20}\right)(4) = 0 - \left(-\frac{1}{20}\right)(9) = \frac{9}{20}.$$

Therefore, the equation of the falling segment is

$$y = m_2 t + b_2 = \frac{9 - t}{20}.$$

The adjacent rules agree at $t = 4$: both give $\frac{1}{4}$.

3.4 Complete pdf

Including the values outside the possible range of commute times, the complete probability density function is

$$f_T(t) = \begin{cases} \dfrac{t - 1}{12}, & 1 \le t \le 4, \\ \dfrac{9 - t}{20}, & 4 < t \le 9, \\ 0, & \text{otherwise.} \end{cases}$$

3.5 Probability of More Than 6 Hours

The apprentice wants to know the chance that the commute takes more than 6 hours. This corresponds to the area under the density from 6 to the maximum possible commute time of 9 hours:

$$P(T > 6) = \frac{1}{2}(9 - 6)\left(f_T(6) + f_T(9)\right).$$

First, evaluate the density at the two endpoints:

$$f_T(6) = \frac{9 - 6}{20} = \frac{3}{20},$$

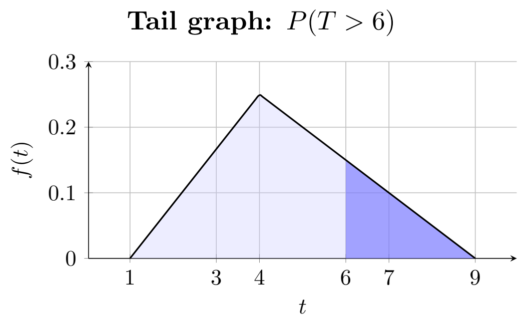
$$f_T(9) = \frac{9 - 9}{20} = 0.$$

Substituting these values gives

$$\begin{aligned} P(T > 6) &= \frac{1}{2}(9 - 6)\left(\frac{3}{20} + 0\right) \\ &= \frac{1}{2}(3)\left(\frac{3}{20}\right) \\ &= \frac{9}{40}. \end{aligned}$$

Thus, the probability of a commute lasting more than 6 hours is

$$\boxed{P(T > 6) = 0.225}.$$



3.6 Probability Between 3 and 7 Hours

The apprentice also wants to know the chance that the commute takes between 3 and 7 hours. We use the complement rule: the event $3 \le T \le 7$ is the complement of the two tail events $T < 3$ and $T > 7$. Therefore,

$$P(3 \le T \le 7) = 1 - P(T < 3) - P(T > 7).$$

The two complementary regions are triangular, so their probabilities are given by

$$P(T < 3) = \frac{1}{2}(3-1)f_T(3), \qquad P(T > 7) = \frac{1}{2}(9-7)f_T(7).$$

Hence,

$$P(3 \le T \le 7) = 1 - \frac{1}{2}(3-1)f_T(3) - \frac{1}{2}(9-7)f_T(7).$$

First, evaluate the density at the two relevant points:

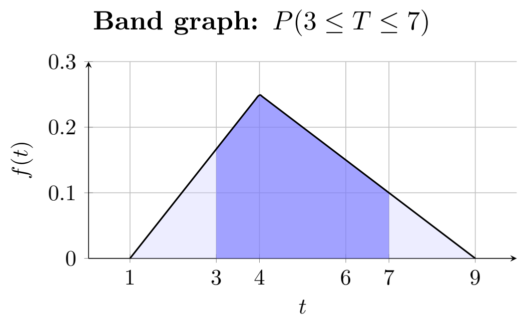
$$\begin{aligned} f_T(3) &= \frac{3-1}{12} = \frac{1}{6}, \\ f_T(7) &= \frac{9-7}{20} = \frac{1}{10}. \end{aligned}$$

Substituting these values gives

$$\begin{aligned} P(3 \le T \le 7) &= 1 - \frac{1}{2}(3-1)\left(\frac{1}{6}\right) - \frac{1}{2}(9-7)\left(\frac{1}{10}\right) \\ &= 1 - \frac{1}{6} - \frac{1}{10} \\ &= \frac{11}{15}. \end{aligned}$$

Thus, the probability of a commute lasting between 3 and 7 hours is

$$\boxed{P(3 \le T \le 7) = \frac{11}{15} \approx 73.3\%}.$$



4 Timing of an Unexpected Market Movement [C3-4]

A financial monitoring system records the time of day at which an unexpected movement in an international currency market occurs. The movement is equally likely to occur at any time during a 24-hour period, so no part of the day is more likely than another.

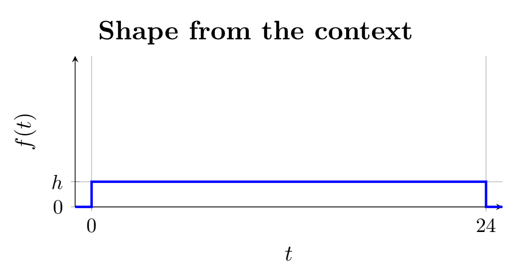
Questions/Tasks.

C3 Determine the complete probability density function for the time at which the market movement occurs. Then determine the probability that the movement occurs between 9 p.m. and 5 a.m. **[4]**

C3 - Uniform probability distribution [4]

4.1 Support and Shape

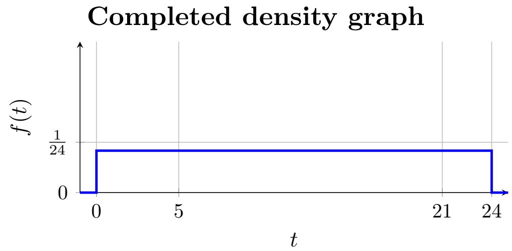
Let T represent the time of day, measured in hours after midnight. The random variable T has support $[0, 24)$. The context says that the market movement is equally likely to occur at any time during the day, so the density is a rectangle of unknown height h and is zero outside the support.



4.2 Total Area = 1

The rectangle must have area 1:

$$1 = (24 - 0)h = 24h, \qquad h = \frac{1}{24}.$$



4.3 Straight Line Rule

The only segment is horizontal. Thus $m = 0$, and its constant rule on the support is

$$f_T(t) = \frac{1}{24};$$

no nontrivial line calculation is needed.

4.4 Complete pdf

Including the values outside the possible range of times, the complete probability density function is

$$f_T(t) = \begin{cases} \frac{1}{24}, & 0 \le t < 24, \\ 0, & \text{otherwise.} \end{cases}$$

4.5 Probability Between 9 p.m. and 5 a.m.

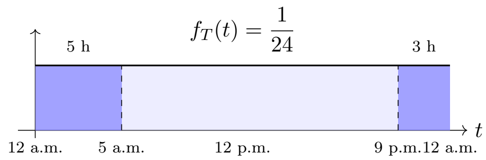
The interval from 9 p.m. to 5 a.m. crosses midnight. Since T is measured in hours after midnight, the relevant interval consists of two parts:

$$21 \le T < 24 \qquad \text{and} \qquad 0 \le T \le 5.$$

The first part covers 3 hours, from 9 p.m. to midnight, while the second part covers 5 hours, from midnight to 5 a.m. Therefore,

$$\begin{aligned} P(\text{between 9 p.m. and 5 a.m.}) &= P(21 \le T < 24) + P(0 \le T \le 5) \\ &= (24 - 21)\frac{1}{24} + (5 - 0)\frac{1}{24} \\ &= \frac{1}{8} + \frac{5}{24} \\ &= \frac{8}{24} \\ &= \boxed{\frac{1}{3} \approx 33.33\%}. \end{aligned}$$

Probability between 9 p.m. and 5 a.m.



5 Direction of an Unexpected Sensor Reading [C3-4]

A directional sensor records the direction in which it is pointing. The sensor can point in any direction around a complete circle, and every direction is equally likely. Let A represent the direction of the sensor, measured in radians from a fixed reference marker at 0 radians.

Questions/Tasks.

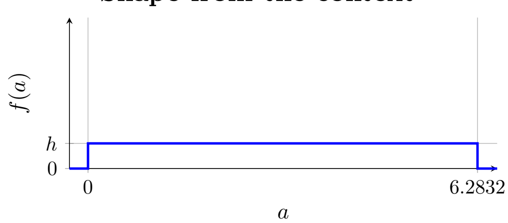
C3 Determine the complete probability density function for the direction of the sensor. Then determine which is more probable: the sensor pointing within $\pi/12$ radians of the reference marker, or the sensor pointing within $10°$ of the reference marker. **[4]**

C3 - Uniform probability distribution [4]

5.1 Support and Shape

Let A represent the direction of the sensor, measured in radians from the reference marker. One complete rotation is represented by the interval $[0, 2\pi)$. The context says that every direction is equally likely, so the density is a rectangle of unknown height h and is zero outside the support.

Shape from the context

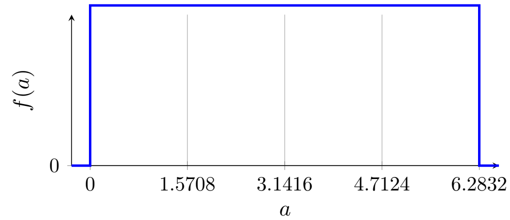


5.2 Total Area = 1

The rectangle must have area 1:

$$1 = (2\pi - 0)h = 2\pi h \qquad h = \frac{1}{2\pi}$$

Completed density graph



5.3 Straight Line Rule

The only segment is horizontal. Thus $m = 0$, and its constant rule on the support is

$$f_A(a) = \frac{1}{2\pi}$$

no nontrivial line calculation is needed.

5.4 Complete pdf

Including the values outside the possible range of directions, the complete probability density function is

$$f_A(a) = \begin{cases} \frac{1}{2\pi} & 0 \le a < 2\pi \\ 0 & \text{otherwise} \end{cases}$$

5.5 Probability Within $\pi/12$ of the Reference Marker

The reference marker is at 0 radians. Being within $\pi/12$ radians of this marker means that the angular distance from 0 is at most $\pi/12$.

Because the support is represented by $[0, 2\pi)$, this region occurs on both sides of the reference marker:

$$0 \le A \le \frac{\pi}{12} \qquad \text{or} \qquad \frac{23\pi}{12} \le A < 2\pi$$

The total angular width is therefore

$$\frac{\pi}{12} + \left(2\pi - \frac{23\pi}{12}\right) = \frac{\pi}{6}$$

Therefore,

$$\begin{aligned} P\left(\text{within } \frac{\pi}{12} \text{ of the reference marker}\right) &= \frac{\pi/6}{2\pi} \\ &= \boxed{\frac{1}{12} \approx 8.33\%} \end{aligned}$$

5.6 Probability Within 10° of the Reference Marker

A complete rotation measures 2π radians, which is equivalent to 360°. The comparison is given in degrees: being within 10° of the reference marker means being within 10° on either side of 0°.

In radians, this angular distance is

$$10^\circ = 10\left(\frac{\pi}{180}\right) = \frac{\pi}{18} \text{ radians.}$$

Thus, the region within 10° of the reference marker has total angular width

$$\text{Degrees:} \qquad 2(10^\circ) = 20^\circ,$$

$$\text{Radians:} \qquad 2\left(\frac{\pi}{18}\right) = \frac{\pi}{9}.$$

The probability can then be calculated using either unit:

$$\begin{aligned} P(\text{within } 10^\circ \text{ of the reference marker}) &= \frac{20^\circ}{360^\circ} = \frac{1}{18}, \\ &= \frac{\pi/9}{2\pi} = \frac{1}{18}. \end{aligned}$$

Therefore,

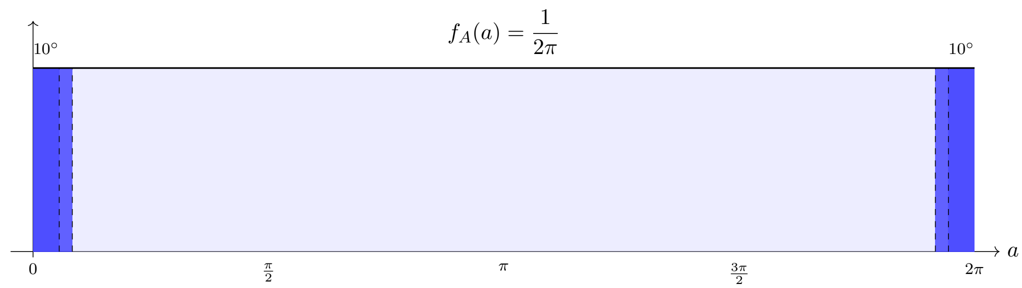
$$\boxed{P(\text{within } 10^\circ \text{ of the reference marker}) = \frac{1}{18} \approx 5.56\%}$$

Since

$$\frac{1}{12} \approx 8.33\% > \frac{1}{18} \approx 5.56\%,$$

$$\boxed{\text{being within } \frac{\pi}{12} \text{ radians of the reference marker is more probable.}}$$

Comparison of the two regions



6 Waiting for a Study-Room Opening [C4-5]

At a university library in Bogotá, study rooms become available at an average rate of one room every 5 minutes.

Questions/Tasks.

C4 Determine the probability that the next study room becomes available more than 7 minutes after the start of the observation. **[4]**

C4 - Exponential probability distribution [4]

The average rate is one available room every 5 minutes. Therefore, the exponential rate parameter is

$$\lambda = \frac{1}{5} = 0.20.$$

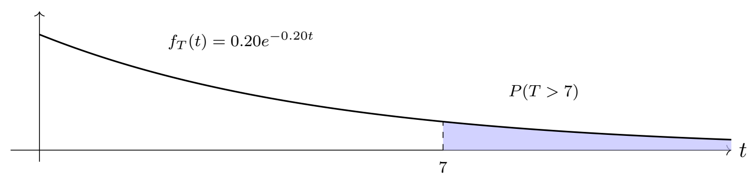
For an exponential random variable, the probability that T is greater than a is

$$P(T > a) = e^{-\lambda a}. \qquad \text{In this problem,} \qquad a = 7, \qquad \lambda = 0.20.$$

$$P(T > 7) = e^{-(0.20)(7)}.$$

$$P(T > 7) = e^{-1.4} \approx 0.2466 = 24.66\%.$$

Probability region



7 Waiting for a Market Price Alert [C4-5]

A financial monitoring system detects significant price changes in a commodity market at an average rate of 3 alerts every 20 minutes.

Questions/Tasks.

C4 Determine the probability that the next significant price change is detected within 6 minutes after the start of the observation. **[4]**

C4 - Exponential probability distribution [4]

The average rate is 3 significant price changes every 20 minutes. Therefore, the exponential rate parameter is

$$\lambda = \frac{3}{20} = 0.15.$$

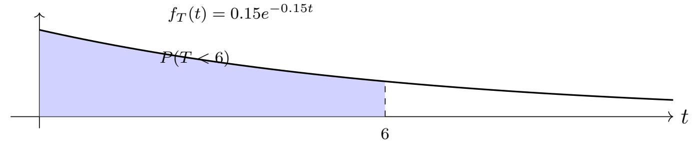
For an exponential random variable, the probability that T is less than a is

$$P(T < a) = 1 - e^{-\lambda a}. \qquad \text{In this problem,} \qquad a = 6, \qquad \lambda = 0.15.$$

$$P(T < 6) = 1 - e^{-(0.15)(6)}.$$

$$P(T < 6) = 1 - e^{-0.9} \approx 0.5934 = 59.34\%.$$

Probability region



8 Waiting for a Demand Alert [C4-5]

An economic forecasting system detects significant changes in consumer demand at an average rate of 4 alerts every 30 minutes.

Questions/Tasks.

C4 Determine the probability that the next significant change in consumer demand is detected between 3 and 8 minutes after the observation time. **[4]**

C4 - Exponential probability distribution [4]

The average rate is 4 significant demand changes every 30 minutes. Therefore, the exponential rate parameter is

$$\lambda = \frac{4}{30} = \frac{2}{15} \approx 0.1333$$

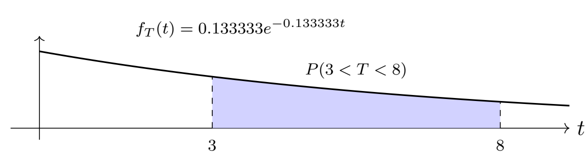
For an exponential random variable, the probability that T lies between a and b is

$$P(a < T < b) = e^{-\lambda a} - e^{-\lambda b} \qquad \text{In this problem,} \qquad a = 3, \qquad b = 8, \qquad \lambda = \frac{2}{15}$$

$$P(3 < T < 8) = e^{-\left(\frac{2}{15}\right)(3)} - e^{-\left(\frac{2}{15}\right)(8)}$$

$$P(3 < T < 8) = e^{-0.4} - e^{-1.0667} \approx 0.3009 = 30.09\%$$

Probability region



9 Processing Investment Transactions [C4-5]

An investment platform processes buy and sell transactions throughout the trading day. The time between consecutive transactions is modeled by an exponential distribution. Based on previous observations, there is a 75% probability that the next transaction is processed within 8 minutes. On average, how many transactions are expected to be processed in a 10-minute interval?

Questions/Tasks.

C5 Determine the average number of transactions expected to be processed in a 10-minute interval. **[5]**

C4.9 - Finding the expected number of transactions

There is a 75% probability that the next transaction is processed within 8 minutes. First, find the rate parameter λ.

General	$P(T < t) = 1 - e^{-\lambda t}$
Context	$P(T < 8) = 0.75$

Therefore,

$$0.75 = 1 - e^{-8\lambda} \quad \longrightarrow \quad e^{-8\lambda} = 0.25$$

$$\longrightarrow \quad \ln(0.25) = -8\lambda \quad \longrightarrow \quad \lambda = -\frac{\ln(0.25)}{8}$$

$$\boxed{\lambda \approx 0.1733 \text{ transactions/minute}}$$

The expected number of transactions in t minutes is

$$E[N(t)] = \lambda t.$$

For a 10-minute interval,

$$E[N(10)] = (0.1733)(10)$$

$$\boxed{E[N(10)] \approx 1.733 \text{ transactions}}$$

10 Waiting for a Loan Approval [C4-5]

A financial institution processes loan applications throughout the day. The processing time for a randomly selected application is modeled by an exponential distribution. Based on previous observations, there is a 30% probability that an application takes more than 12 minutes to process. Find the average number of applications processed per minute.

Questions/Tasks.

C5 Determine the average number of applications processed per minute. **[5]**

C4.9 - Finding the rate parameter

There is a 30% probability that an application takes more than 12 minutes to process. Find λ.

General	$P(T > t) = e^{-\lambda t}$
Context	$P(T > 12) = 0.30$

Therefore,

$$0.30 = e^{-12\lambda}$$

$$\longrightarrow \quad \ln(0.30) = -12\lambda \quad \longrightarrow \quad \lambda = -\frac{\ln(0.30)}{12}$$

$$\boxed{\lambda \approx 0.1004 \text{ applications/minute}}$$

11 Comparing Investment Transaction Rates [C4-10]

Two investment platforms process buy and sell transactions throughout the trading day. The transactions processed by each platform are modeled by independent Poisson processes.

For Platform A, there is a 75% probability that the next transaction is processed within 8 minutes. For Platform B, there is a 60% probability that the next transaction is processed within 5 minutes.

Assume both platforms begin processing transactions at the same time. On average, how many minutes does it take for one platform to be ahead of the other by 1 transaction?

Questions/Tasks.

C10 Determine the rate parameter λ for each platform and the expected time until one platform is ahead of the other by 1 transaction. **[10]**

C4.10 - Comparing exponential rates

Determine the rate parameters for the two platforms.

For Platform A,

$$P(T < 8) = 0.75.$$

Using

$$P(T < t) = 1 - e^{-\lambda t},$$

we obtain

$$0.75 = 1 - e^{-8\lambda_A} \quad \longrightarrow \quad e^{-8\lambda_A} = 0.25.$$

Therefore,

$$\lambda_A = -\frac{\ln(0.25)}{8} \approx 0.1733 \text{ transactions/minute}.$$

For Platform B,

$$P(T < 5) = 0.60.$$

Thus,

$$0.60 = 1 - e^{-5\lambda_B} \quad \longrightarrow \quad e^{-5\lambda_B} = 0.40.$$

Therefore,

$$\lambda_B = -\frac{\ln(0.40)}{5} \approx 0.1833 \text{ transactions/minute}.$$

Platform B has the higher rate, with a rate difference of

$$\lambda_B - \lambda_A \approx 0.1833 - 0.1733 = 0.0100 \text{ transactions/minute}.$$

The expected difference in the number of transactions after t minutes is

$$E[N_B(t) - N_A(t)] = (\lambda_B - \lambda_A)t.$$

To find the time at which Platform B is expected to be ahead by 1 transaction, set

$$(\lambda_B - \lambda_A)t = 1.$$

Therefore,

$$t = \frac{1}{\lambda_B - \lambda_A} = \frac{1}{0.0100} = 100 \text{ minutes.}$$

Platform B is expected to be ahead by 1 transaction after approximately 100 minutes.